Efficiency Improvement of Deniable FHE: Tighter Deniability Analysis and TFHE-based Construction

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Abstract. Fully homomorphic encryption (FHE) is a cryptographic scheme that can take ciphertexts as inputs and compute a new ciphertext of a function of the underlying messages without decryption. FHE has been attracting attention along with the growing interest in privacy-preserving technologies. In terms of privacy-preserving technology, deniable encryption is also important. Deniable encryption enables a user, who may be forced to reveal the messages corresponding to the user's public ciphertexts, to lie about which messages the user encrypted. Agrawal et al. (CRYPTO 2021) introduced deniable FHE (DFHE) that combines FHE with deniable encryption, and proposed a transformation from an FHE scheme that satisfies specific special requirements, called special FHE, to a DFHE scheme. They also showed a construction of a special FHE scheme based on the BGV (Brakerski–Gentry–Vaikuntanathan) scheme. However, in the construction by Agrawal et al., one must store all the extensive randomness used for encryption in order to lie, and a bootstrapping operation, which takes a long time to execute, is a bottleneck in execution speed. In this paper, we show that by providing a tighter upper bound on deniability, we can reduce the size of the stored randomness and the required number of bootstrapping in the construction by Agrawal et al. In addition, we show that TFHE (Chillotti et al., J. Cryptol., 2020; Joye, CT-RSA 2024), which is known as a FHE scheme with fast bootstrapping, satisfies the requirements of special FHE, and thus can realize a faster DFHE scheme than the BGV-based construction.

Keywords: Deniable fully homomorphic encryption · Deniability · TFHE.

1 Introduction

1.1 Background

In recent years, there has been a growing interest in privacy-preserving data utilization. Consequently, research on techniques for secure cross-organizational data collaboration and analysis of sensitive information has become an active field. In this context, secure computation, a class of cryptographic techniques that enables computation on private inputs without revealing them, is attracting significant attention. Multiparty computation [26,17] and homomorphic encryption [23] are primary examples of techniques used to realize secure computation.

Homomorphic encryption is a cryptographic scheme that enables computation on messages by performing certain operations on ciphertexts without decryption. In particular, a scheme that can perform both additions and multiplications for an arbitrary number of times is called fully homomorphic encryption (FHE) [15]. Since the first FHE construction was proposed by Gentry [15,14], various FHE schemes have been proposed, such as BGV [6], B/FV [5,13], GSW [16], DM (or FHEW) [12], CGGI (or TFHE) [11], and CKKS [10,9].

In the context of privacy-preserving technologies, one relevant cryptographic primitive is deniable encryption, proposed by Canetti et al. [8]. Deniable encryption is a cryptographic scheme that enables a user, who may be coerced to reveal the messages corresponding to the user's public ciphertexts, to lie about which messages the user encrypted. This property is useful in scenarios such as electronic voting and cloud computing, where computations are performed on encrypted data provided by other parties.

Agrawal et al. [2] proposed deniable FHE (DFHE), a cryptographic scheme that satisfies the properties of both FHE and deniable encryption, and provided its security model and a construction. They proposed a concrete construction of DFHE by first showing a generic method to convert any FHE scheme that satisfies certain properties (which they term a special FHE) into a DFHE scheme. They then demonstrated how to construct such a special FHE based on the BGV scheme [6].

However, the DFHE construction proposed by Agrawal et al. [2] suffers from two drawbacks in terms of efficiency. The first is that in order to lie about a one-bit message, one must store all $\mathcal{O}\Big(\lambda \cdot \mathsf{poly}(\lambda)^2 \cdot \delta(\lambda)^2 \cdot (\ell_c' + \ell)\Big)$ bits of the randomness used for encryption, where λ is the security parameter, $\mathsf{poly}(\lambda)$ is a positive polynomial, $\delta(\lambda)$ is a parameter for deniability, ℓ_c' is the bit length of elements in ciphertext space, and ℓ is a parameter for encryption. The second is the heavy use of bootstrapping operations, which plays a crucial role in existing FHE constructions but is computationally expensive. Indeed, Agrawal et al.'s construction [2] requires $\mathcal{O}\Big(\lambda \cdot \mathsf{poly}(\lambda)^2 \cdot \delta(\lambda)^2\Big)$ bootstrapping operations to encrypt a one-bit message, where λ is the security parameter, $\mathsf{poly}(\lambda)$ is a positive polynomial, and $\delta(\lambda)$ is a parameter for deniability. Consequently, the overall performance of the DFHE scheme is significantly impacted by the speed of a single bootstrapping operation.

1.2 Our Contributions

In this paper, we propose methods to mitigate two efficiency drawbacks in Agrawal et al.'s DFHE scheme [2]: the large randomness size and the performance overhead caused by the heavy use of bootstrapping.

First, in Section 4, we provide a tighter upper bound on deniability of Agrawal et al.'s DFHE scheme [2]. Since the bit length of the randomness and the number of bootstrapping operations in the encryption algorithm depend on the deniability, these values can be reduced by achieving a tighter upper bound on deniability. By providing the tighter bound, both the bit length of the randomness and the number of bootstrapping operations can be reduced by 84%.

Second, in Section 5, we show that TFHE [11,20], which is known as a FHE scheme with fast bootstrapping, satisfies the requirements of special FHE, and thus can realize a faster DFHE scheme than the BGV-based construction.

2 Preliminaries

2.1 Notations

Throughout the paper, we use $\lambda \in \mathbb{N}$ to denote the security parameter. We write $y \leftarrow A(x)$ to denote that the probabilistic polynomial-time (PPT) algorithm A takes x as input and outputs y. To explicitly show that A uses randomness r, we write $y \leftarrow A(x; r)$. For a distribution D, we denote by $d \leftarrow D$ the operation of sampling d according to D. For a finite set S, we denote by $s \xleftarrow{\$} S$ the operation of sampling s uniformly at random from S. For two integers $a, b \in \mathbb{Z}_{\geq 0}$ with $a \leq b$, we write $[a, b] \coloneqq \{a, a+1, \ldots, b\}$ and $[b] \coloneqq [1, b]$. For a real number $x \in \mathbb{R}$, we denote by $\lfloor x \rceil$ the closest integer to x. For integers $a \in \mathbb{Z}$ and $b \in \mathbb{Z} \setminus \{0\}$, we use $b|a$ to denote that b divides a. $\varphi(\cdot)$ denotes Euler's totient function. We use bold lowercase letters (e.g., $\boldsymbol{v}$) to denote vectors. For two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ of the same dimension, we denote by $\langle \boldsymbol{a}, \boldsymbol{b} \rangle$ the inner product of $\boldsymbol{a}$ and $\boldsymbol{b}$. For a bit string $\boldsymbol{x} = (x_1, \ldots, x_n) \in \{0,1\}^n$, we define the multiplicity of 1 as $M_1(\boldsymbol{x}) \coloneqq \#\{i \in [n] \mid x_i = 1\}$. We use $\mathsf{poly}(\lambda)$ and $\mathsf{negl}(\lambda)$ to denote a positive polynomial function of λ and a negligible function of λ, respectively.

2.2 Useful Lemmas

We will use the following lemmas throughout this paper.

Lemma 1. *For all $n \in \mathbb{N}$ and all $k \in [0, n]$, we have*

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k} \quad \textit{and} \quad \binom{n}{k+1} = \frac{n-k}{k+1}\binom{n}{k}.$$

Lemma 2 ([24]). *For all $n \in \mathbb{N}$, we have*

$$\sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}.$$

Lemma 3 ([4]). *For all $n \in \mathbb{N}$, we have*

$$\left(1+\frac{1}{n}\right)^n < e.$$

2.3 Fully Homomorphic Encryption

We recall the definition of FHE.

Definition 1 (Fully Homomorphic Encryption). *A public-key fully homomorphic encryption scheme* FHE *for a message space* $\mathcal{M}$ *consists of PPT algorithms* $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *with the following syntax:*

- $\mathsf{Gen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$*: On input the unary representation of the security parameter* λ*, the key-generation algorithm* Gen *outputs a public key* pk *and secret key* sk*.*
- $\mathsf{Enc}(\mathsf{pk}, m) \to \mathsf{ct}$*: On input a public key* pk *and a message* m*, the encryption algorithm* Enc *outputs a ciphertext* ct*.*
- $\mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k) \to \mathsf{ct}$*: On input a public key* pk*, a circuit* $C\colon \mathcal{M}^k \to \mathcal{M}$*, and ciphertexts* $\mathsf{ct}_1, \ldots, \mathsf{ct}_k$*, the evaluation algorithm* Eval *outputs a ciphertext* ct*.*
- $\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}) \to m$*: On input a secret key* sk *and a ciphertext* ct*, the decryption algorithm* Dec *outputs a message* $m \in \mathcal{M}$*.*

We omit the definitions of correctness, compactness, and IND-CPA security of FHE scheme here. For the formal definitions of them, see Definitions 6, 7 and 8 in Appendix A.1.

In FHE, performing operations on ciphertexts increases the amount of noise they contain. This noise growth limits the number of homomorphic operations that can be performed on a ciphertext. A technique to reduce this noise without decrypting the ciphertext was introduced by Gentry [15,14] and is known as bootstrapping. All currently known FHE schemes are constructed based on this bootstrapping technique. The definition of bootstrapping is as follows.

Definition 2 (Bootstrapping Procedure). *Let* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *be a public-key FHE scheme for a message space* $\mathcal{M}$ *with a ciphertext space* $\mathcal{R}^{\ell_c}$*. The bootstrapping procedure, denoted by* $\mathsf{boot}\colon \mathcal{R}^{\ell_c} \to \mathcal{R}^{\ell_c}$*, is defined as*

$$\mathsf{boot}(x) = \mathsf{Eval}(\mathsf{pk}, \mathsf{Dec}_x, \mathsf{ct}_{\mathsf{sk}}),$$

where $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$*,* $\mathsf{ct}_{\mathsf{sk}} \leftarrow \mathsf{Enc}(\mathsf{pk}, \mathsf{sk})$*, and* $\mathsf{Dec}_x(\mathsf{sk}) = \mathsf{Dec}(\mathsf{sk}, x)$*.*

2.4 Deniable Fully Homomorphic Encryption

We recall the definition of DFHE.

Definition 3 (Deniable FHE [2]). *A public-key deniable fully homomorphic encryption scheme* DFHE *for a message space* $\mathcal{M}$ *consists of PPT algorithms* $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ *with the following syntax:*

- $\mathsf{Gen}(1^\lambda) \to (\mathsf{dpk}, \mathsf{dsk})$*: On input the unary representation of the security parameter* λ*, the key-generation algorithm* Gen *outputs a public key* dpk *and secret key* dsk*.*
- $\mathsf{Enc}(\mathsf{dpk}, m; r) \to \mathsf{dct}$*: On input a public key* dpk *and a message* m*, the encryption algorithm* Enc *uses an* ℓ*-bit randomness* r *and outputs a ciphertext* dct*.*
- $\mathsf{Eval}(\mathsf{dpk}, C, \mathsf{dct}_1, \ldots, \mathsf{dct}_k) \to \mathsf{dct}$*: On input a public key* dpk*, a circuit* $C\colon \mathcal{M}^k \to \mathcal{M}$*, and ciphertexts* $\mathsf{dct}_1, \ldots, \mathsf{dct}_k$*, the evaluation algorithm* Eval *outputs a ciphertext* dct*.*
- $\mathsf{Dec}(\mathsf{dsk}, \mathsf{dct}) \to m$*: On input a secret key* dsk *and a ciphertext* dct*, the decryption algorithm* Dec *outputs a message* $m \in \mathcal{M}$*.*
- $\mathsf{Fake}(\mathsf{dpk}, m, r, m^*) \to r^*$*: On input a public key* dpk*, an original message* m*, an original* ℓ*-bit randomness* r*, and a fake message* $m^* \in \mathcal{M}$*, the fake algorithm* Fake *outputs a fake* ℓ*-bit randomness* r^**.*

We say that a DFHE scheme $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ is correct, compact, and IND-CPA secure if the scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies correctness, compactness, and IND-CPA security of FHE, as in Definitions 6, 7 and 8, respectively. The notions of deniability and deniability compactness of DFHE were defined by Agrawal et al. [2]. Deniability is defined as follows.

Definition 4 (Deniability [2]). *A DFHE scheme* $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ *is* $1/\delta(\lambda)$*-deniable if for all PPT adversaries* $\mathcal{A}$*, we have*

$$\left|\Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1\right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1\right]\right| \leq \frac{1}{\delta(\lambda)},$$

where $\mathsf{DenGame}^b_{\mathcal{A}}$ *is a game between a challenger and a PPT adversary* $\mathcal{A}$ *with a challenge bit* b *defined as follows:*

- *The challenger samples* $(\mathsf{dpk}, \mathsf{dsk}) \leftarrow \mathsf{Gen}(1^\lambda)$*, and sends* dpk *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *chooses* $m, m^* \in \mathcal{M}$*, and sends* (m, m^*) *to the challenger.*
- *The challenger samples* $r \xleftarrow{\$} \{0,1\}^\ell$ *and computes* $r^* \leftarrow \mathsf{Fake}(\mathsf{dpk}, m, r, m^*)$*; if* $b = 0$ *sends* $(m^*, r, \mathsf{Enc}(\mathsf{dpk}, m^*; r))$ *to* $\mathcal{A}$*, else if* $b = 1$*, sends* $(m^*, r^*, \mathsf{Enc}(\mathsf{dpk}, m; r))$ *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *outputs a bit* b' *which we define as the output of the game.*

We omit the definition of deniability compactness here. For the formal definition of deniability compactness, see Definition 12 in Appendix A.2.

3 Agrawal et al.'s DFHE Scheme

The DFHE constructions proposed by Agrawal et al. [2] rely on an FHE scheme, called special FHE, that satisfies some special properties. They proposed two variants of special FHE schemes in the full version [1]: a strong version and a weak version. In this paper, we employ the latter, defined as follows.

Definition 5 (Special FHE (Weak Version) [1]). *An FHE scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *for the message space* $\mathcal{M} = \{0,1\}$ *with ciphertext space* $\mathcal{R}^{\ell_c}$ *is a special FHE scheme if* FHE *satisfies the following additional properties:*

1. *The evaluation algorithm* Eval *and the decryption algorithm* Dec *are deterministic.*
2. *The distribution* $\mathsf{Enc}(\mathsf{pk}, m; U^\ell)$ *is computationally indistinguishable from* $\mathcal{R}^{\ell_c}$*, where* U^ℓ *is a uniform distribution over* $\{0,1\}^\ell$*,* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$*, and* $m \in \mathcal{M}$*. Moreover, the distribution* $\mathsf{boot}(\mathcal{R}^{\ell_c})$ *is computationally indistinguishable from* $\mathcal{R}^{\ell_c}$*.*
3. *The decryption algorithm* Dec *always outputs a message from the message space* $\mathcal{M}$*.*
4. *On input a secret key* sk *and random element in the ciphertext space* $\mathcal{R}^{\ell_c}$*, the decryption algorithm* Dec *outputs* 0 *with non-negligible probability.*
5. *The scheme* FHE *is circular secure.*[1]

We recall Agrawal et al.'s DFHE scheme [1, Section 7.1] below.

Construction 1 (DFHE [1]) *Let a scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *be a special FHE scheme for the message space* $\mathcal{M} = \{0,1\}$ *with ciphertext space* $\mathcal{R}^{\ell_c}$*. We let* $n = 100\delta(\lambda)^2$ *and* $k' = \lambda(\mathsf{poly}(\lambda))^2$*. Let* $\oplus_2 \colon (\mathcal{R}^{\ell_c})^n \to \mathcal{R}^{\ell_c}$ *be a homomorphic evaluation of addition modulo 2.*[2] *A compact public-key* $1/\delta(\lambda)$*-deniable FHE scheme for the message space* $\mathcal{M} = \{0,1\}$*,* $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$*, is described below.*

- $\mathsf{DFHE.Gen}(1^\lambda) \to (\mathsf{dpk}, \mathsf{dsk})$*: On input the unary representation of the security parameter* λ*, do the following:*
  1. $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{FHE.Gen}(1^\lambda)$*.*
  2. $\mathsf{ct_{sk}} \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, \mathsf{sk})$*.*
  3. *Output* $\mathsf{dpk} = (\mathsf{pk}, \mathsf{ct_{sk}})$ *and* $\mathsf{dsk} = \mathsf{sk}$*.*
- $\mathsf{DFHE.Enc}(\mathsf{dpk}, m) \to \mathsf{dct}$*: On input a public key* dpk *and a message* m*, do the following:*
  1. *Parse* $\mathsf{dpk} \coloneqq (\mathsf{pk}, \mathsf{ct_{sk}})$*.*
  2. $x_1, \ldots, x_n \xleftarrow{\$} \{0,1\}$ *s.t.* $\sum_{i=1}^n x_i = m \pmod 2$*.*
  3. *For all* $i \in [n]$*, if* $x_i = 0$*, select* R_i *as follows:*

[1] For the formal definition of circular security [7], see Definition 9 in Appendix A.1.

[2] For the formal definition of $\oplus_2$, see Definition 11 in Appendix A.1.

(a) *For all* $j \in [k']$, $A_{i,j} \xleftarrow{\$} \mathcal{R}^{\ell_c}$ *and set* $T_{i,j} = \mathsf{boot}(A_{i,j})$.
(b) *Set* $R_i = \mathsf{FHE.Eval}(\mathsf{pk}, \mathsf{AND}, T_{i,1}, \ldots, T_{i,k'})$.

4. *For all* $i \in [n]$, *if* $x_i = 1$, *select* R_i *as follows:*
   (a) *For all* $j \in [k']$, $r_{i,j} \xleftarrow{\$} \{0,1\}^{\ell}$, $A_{i,j} \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, 1; r_{i,j})$, *and set* $T_{i,j} = \mathsf{boot}(A_{i,j})$.
   (b) *Set* $R_i = \mathsf{FHE.Eval}(\mathsf{pk}, \mathsf{AND}, T_{i,1}, \ldots, T_{i,k'})$.
5. *Output* $\mathsf{dct} = \oplus_2(\mathsf{boot}(R_1), \ldots, \mathsf{boot}(R_n))$.

– $\mathsf{DFHE.Eval}(\mathsf{dpk}, C, \mathsf{dct}_1, \ldots, \mathsf{dct}_k) \to \mathsf{dct}$: *On input a public key* dpk, *a circuit* $C\colon \mathcal{M}^k \to \mathcal{M}$, *and ciphertexts* $\mathsf{dct}_1, \ldots, \mathsf{dct}_k$, *do the following:*
  1. *For all* $i \in [k]$, *interpret* dct_i *as a* FHE *ciphertext* ct_i.
  2. *Output* $\mathsf{dct} = \mathsf{FHE.Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$.
– $\mathsf{DFHE.Dec}(\mathsf{dsk}, \mathsf{dct}) \to m$: *On input a secret key* dsk *and a ciphertext* dct, *do the following:*
  1. *Interpret* dsk *and* dct *as* FHE *secret key* sk *and* FHE *ciphertext* ct.
  2. *Output* $\mathsf{FHE.Dec}(\mathsf{sk}, \mathsf{ct})$.
– $\mathsf{DFHE.Fake}(\mathsf{dpk}, m, \mathsf{rand}, m^*) \to \mathsf{rand}^*$: *On input a public key* dpk, *an original message* m, *an original* ℓ*-bit randomness* rand, *and a fake message* $m^* \in \mathcal{M}$, *do the following:*
  1. *If* $m = m^*$, *output* $\mathsf{rand}^* = \mathsf{rand}$.
  2. *Parse* $\mathsf{dpk} \coloneqq (\mathsf{pk}, \mathsf{ct}_{\mathsf{sk}})$ *and* $\mathsf{rand} = (x_1, \ldots, x_n, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_n)$, *where* $x_1, \ldots, x_n \in \{0,1\}$, *and for all* $i \in [n]$, *if* $x_i = 1$, *then* $\boldsymbol{r}_i = (r_{i,1}, \ldots, r_{i,k'})$; *else if* $x_i = 0$, *then* $\boldsymbol{r}_i = (A_{i,1}, \ldots, A_{i,k'})$.
  3. $i^* \xleftarrow{\$} [n]$ *s.t.* $x_{i^*} = 1$. *If there is no such* i^*, *output "cheating impossible"; else:*
     (a) *For all* $j \in [k']$, $A_{i^*,j} \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, 1; r_{i^*,j})$.
     (b) *Set* $x^*_{i^*} = 0$ *and* $\boldsymbol{r}^*_{i^*} = (A_{i^*,1}, \ldots, A_{i^*,k'})$.
     (c) *For all* $i \in [n] \setminus \{i^*\}$, *set* $x^*_i = x_i$ *and* $\boldsymbol{r}^*_i = \boldsymbol{r}_i$.
  4. *Output* $\mathsf{rand}^* = (x^*_1, \ldots, x^*_n, \boldsymbol{r}^*_1, \ldots, \boldsymbol{r}^*_n)$.

Regarding the deniability of the DFHE scheme (Construction 1), Proposition 1 holds as follows.

Proposition 1 ([1]). *For all PPT adversaries* $\mathcal{A}$, *we have*

$$\left|\Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1\right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1\right]\right| \le \frac{1}{2^n}\left\{\sum_{k \ge n/2} \binom{n}{k}\left(1 - \frac{n-k}{k+1}\right) + \sum_{k < n/2} \binom{n}{k}\left(\frac{n-k}{k+1} - 1\right)\right\} \tag{1}$$

$$\le \frac{10}{\sqrt{n}}. \tag{2}$$

If we let $n = 100\delta(\lambda)^2$, $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ *is* $1/\delta(\lambda)$*-deniable.*

We here emphasize that the parameter n drastically affects the efficiency of the above DFHE scheme, since the randomness size and the number of bootstrapping operations depend on n. Let us look into it below.

Let $\boldsymbol{x} = (x_1, \ldots, x_n)$. We denote by ℓ'_c the bit length of elements in the ciphertext space $\mathcal{R}^{\ell_c}$ (that is, $\ell'_c = \lceil \ell_c \log_2(|\mathcal{R}|) \rceil$). The bit length of rand is $n + k'\ell'_c(n - M_1(\boldsymbol{x})) + k'\ell M_1(\boldsymbol{x})$. Since we can set $M_1(\boldsymbol{x}) = cn$ for $0 \le c \le 1$, the bit length of rand is as follows:

$$\underbrace{n}_{\boldsymbol{x}} + \underbrace{k'\ell'_c(n - M_1(\boldsymbol{x}))}_{(A_{i,j})\text{ in Step 3}} + \underbrace{k'\ell M_1(\boldsymbol{x})}_{(r_{i,j})\text{ in Step 4}} = (1 + (1-c)k'\ell'_c + ck'\ell)n. \tag{3}$$

Since both addition modulo 2 and an AND gate can be evaluated homomorphically with a single bootstrapping operation, the total number of bootstrapping operations in the encryption algorithm is as follows:

$$\underbrace{(k' + (k'-1))(n - M_1(\boldsymbol{x}))}_{\text{Step 3}} + \underbrace{(k' + (k'-1))M_1(\boldsymbol{x})}_{\text{Step 4}} + \underbrace{n + (n-1)}_{\text{Step 5}} = (2k'+1)n - 1. \tag{4}$$

4 Tighter Upper Bound on Deniability

In this section, we show that the bit length of the randomness and the number of bootstrapping operations in the encryption algorithm can be reduced by providing a tighter upper bound on deniability of Construction 1. In Section 4.1, we revisit the analysis of the deniability provided by Agrawal et al. [1]. In Section 4.2, we analyze how our tighter deniability bound reduces the bit length of the randomness and the number of bootstrapping operations in encryption algorithm.

4.1 Deniability Analysis

We provide a tighter bound for the upper bound given in Equation (2) of Proposition 1. Specifically, we prove the following Theorem 1.

Theorem 1. *For all PPT adversaries $\mathcal{A}$, we have*

$$\left|\Pr\left[\mathsf{DenGame}_{\mathcal{A}}^{0} = 1\right] - \Pr\left[\mathsf{DenGame}_{\mathcal{A}}^{1} = 1\right]\right| \leq \frac{4}{\sqrt{n}}.$$

If we let $n = 16\delta(\lambda)^2$, the DFHE scheme $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ is $1/\delta(\lambda)$-deniable.

Proof. We have

$$\sum_{k \geq n/2} \binom{n}{k}\left(1 - \frac{n-k}{k+1}\right) \overset{\text{(a)}}{=} \sum_{k=\lceil n/2 \rceil}^{n} \binom{n}{k} - \sum_{k=\lceil n/2 \rceil}^{n} \binom{n}{k+1} \overset{\text{(b)}}{=} \binom{n}{\lceil \frac{n}{2} \rceil} \tag{5}$$

and

$$\sum_{k < n/2} \binom{n}{k}\left(\frac{n-k}{k+1} - 1\right) \overset{\text{(a)}}{=} \sum_{k<n/2} \binom{n}{k+1} - \sum_{k<n/2} \binom{n}{k}. \tag{6}$$

Equation (a) follows from Lemma 1, and Equation (b) follow from $\binom{n}{k} = 0$ when $n < k$.

In addition, when n is an even number, we have

$$\begin{aligned}
\sum_{k<n/2} \binom{n}{k+1} - \sum_{k<n/2} \binom{n}{k} &= \sum_{k=0}^{\frac{n}{2}-1} \binom{n}{k+1} - \sum_{k=0}^{\frac{n}{2}-1} \binom{n}{k} \\
&= \binom{n}{\frac{n}{2}} - \binom{n}{0} \\
&= \binom{n}{\lceil \frac{n}{2} \rceil} - 1 \\
&\overset{\text{(c)}}{=} \frac{n - \lceil \frac{n}{2} \rceil + 1}{\lceil \frac{n}{2} \rceil} \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1 \\
&= \left(1 + \frac{2}{n}\right) \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1
\end{aligned} \tag{7}$$

and when n is an odd number, we have

$$\begin{aligned}
\sum_{k<n/2} \binom{n}{k+1} - \sum_{k<n/2} \binom{n}{k} &= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{k+1} - \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{k} \\
&= \binom{n}{\lfloor \frac{n}{2} \rfloor + 1} - 1 \\
&= \binom{n}{n - (\lfloor \frac{n}{2} \rfloor + 1)} - 1
\end{aligned}$$

$$\overset{\text{(d)}}{=} \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1. \tag{8}$$

Equation (c) follows from Lemma 1, and Equation (d) follows from $\lfloor \frac{n}{2} \rfloor + \lceil \frac{n}{2} \rceil = n$ for all $n \in \mathbb{Z}$.

Hence, according to Equations (6), (7) and (8), we have

$$\sum_{k<n/2} \binom{n}{k} \left(\frac{n-k}{k+1} - 1 \right) \leq \left(1 + \frac{2}{n} \right) \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1. \tag{9}$$

Therefore, according to Equation (1) in Proposition 1, we have

$$\begin{aligned}
\left| \Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1 \right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1 \right] \right| &\leq \frac{1}{2^n} \left\{ \sum_{k \geq n/2} \binom{n}{k} \left(1 - \frac{n-k}{k+1} \right) + \sum_{k<n/2} \binom{n}{k} \left(\frac{n-k}{k+1} - 1 \right) \right\} \\
&\overset{\text{(e)}}{\leq} \frac{1}{2^n} \left\{ \binom{n}{\lceil \frac{n}{2} \rceil} + \left(1 + \frac{2}{n} \right) \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1 \right\} \\
&= \frac{1}{2^n} \left\{ \binom{n}{\lceil \frac{n}{2} \rceil} + \binom{n}{\lceil \frac{n}{2} \rceil - 1} + \frac{2}{n} \binom{n}{\lceil \frac{n}{2} \rceil - 1} - 1 \right\} \\
&\overset{\text{(f)}}{=} \frac{1}{2^n} \binom{n+1}{\lceil \frac{n}{2} \rceil} + \frac{2}{n2^n} \binom{n}{\lceil \frac{n}{2} \rceil - 1} - \frac{1}{2^n} \\
&< \frac{1}{2^n} \binom{n+1}{\lceil \frac{n}{2} \rceil} + \frac{2}{n2^n} \binom{n}{\lceil \frac{n}{2} \rceil - 1} \\
&= \frac{1}{2^n} \binom{n+1}{\lceil \frac{n}{2} \rceil} + \frac{2}{n2^n} \left\{ \binom{n}{\lceil \frac{n}{2} \rceil - 1} + \binom{n}{\lceil \frac{n}{2} \rceil} - \binom{n}{\lceil \frac{n}{2} \rceil} \right\} \\
&\overset{\text{(f)}}{=} \frac{1}{2^n} \binom{n+1}{\lceil \frac{n}{2} \rceil} + \frac{2}{n2^n} \binom{n+1}{\lceil \frac{n}{2} \rceil} - \frac{2}{n2^n} \binom{n}{\lceil \frac{n}{2} \rceil} \\
&< \frac{1}{2^n} \left(1 + \frac{2}{n} \right) \binom{n+1}{\lceil \frac{n}{2} \rceil}.
\end{aligned} \tag{10}$$

Inequality (e) follows from Equations (5) and (9), and Equation (f) follows from Lemma 1.

Furthermore, according to Equations (7) and (8), the equality in Equation (9) holds when n is an even number. Hence, according to Equation (10), we have

$$\left| \Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1 \right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1 \right] \right| < \frac{1}{2^n} \left(1 + \frac{2}{n} \right) \binom{n+1}{\frac{n}{2}}. \tag{11}$$

Moreover, we have

$$\begin{aligned}
\binom{n+1}{\frac{n}{2}} &= \frac{(n+1)!}{\frac{n}{2}!\,\left(n+1-\frac{n}{2}\right)!} = \frac{(n+1)!}{\frac{n}{2}!\,\left(\frac{n}{2}+1\right)!} \\
&\overset{\text{(g)}}{<} \frac{\sqrt{2\pi(n+1)}\left(\frac{n+1}{e}\right)^{n+1} e^{\frac{1}{12(n+1)}}}{\sqrt{2\pi\frac{n}{2}}\left(\frac{n}{2e}\right)^{\frac{n}{2}} e^{\frac{1}{12\cdot\frac{n}{2}+1}} \cdot \sqrt{2\pi\left(\frac{n}{2}+1\right)}\left(\frac{\frac{n}{2}+1}{e}\right)^{\frac{n}{2}+1} e^{\frac{1}{12\left(\frac{n}{2}+1\right)+1}}} \\
&= \sqrt{\frac{n+1}{n\pi\left(\frac{n}{2}+1\right)}}\, \frac{(n+1)^{n+1}}{\left(\frac{n}{2}\right)^{\frac{n}{2}}\left(\frac{n}{2}+1\right)^{\frac{n}{2}+1}} \cdot e^{-\frac{108n^2+228n+155}{12(n+1)(6n+1)(6n+13)}} \\
&\overset{\text{(h)}}{<} \sqrt{\frac{n+1}{n\pi\left(\frac{n}{2}+1\right)}}\, \frac{(n+1)^{n+1}e^0}{\left(\frac{n}{2}\right)^{\frac{n}{2}}\left(\frac{n}{2}+1\right)^{\frac{n}{2}+1}} \\
&= \frac{2\sqrt{2} \cdot 2^n (n+1)^{n+\frac{3}{2}}}{\sqrt{\pi} n^{\frac{n+1}{2}} (n+2)^{\frac{n+3}{2}}}.
\end{aligned} \tag{12}$$

Inequality (g) follows from Lemma 2, and Inequality (h) follows from $f(n) < 0$ for all $n \in \mathbb{N}$ if we let $f(n) \coloneqq -\frac{108n^2+228n+155}{12(n+1)(6n+1)(6n+13)}$.

Therefore, according to Equation (11), we have

$$\begin{aligned}
\left|\Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1\right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1\right]\right| &< \frac{1}{2^n}\left(1+\frac{2}{n}\right)\binom{n+1}{\frac{n}{2}} \\
&\overset{\text{(i)}}{<} \frac{1}{2^n}\cdot\frac{n+2}{n}\cdot\frac{2\sqrt{2}\cdot 2^n(n+1)^{n+\frac{3}{2}}}{\sqrt{\pi}n^{\frac{n+1}{2}}(n+2)^{\frac{n+3}{2}}} \\
&= \frac{2\sqrt{2}}{\sqrt{\pi}}\cdot\frac{(n+1)^{n+\frac{3}{2}}}{n^{\frac{n+3}{2}}(n+2)^{\frac{n+1}{2}}} \\
&< \frac{2\sqrt{2}}{\sqrt{\pi}}\cdot\frac{(n+1)^{n+\frac{3}{2}}}{n^{\frac{n+3}{2}}(n+1)^{\frac{n+1}{2}}} \\
&= \frac{2\sqrt{2}}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{n}}\left(1+\frac{1}{n}\right)\sqrt{\left(1+\frac{1}{n}\right)^n} \\
&\overset{\text{(j)}}{<} \frac{2\sqrt{2}}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{n}}\cdot\frac{3}{2}\cdot\sqrt{e} \\
&= \sqrt{\frac{18e}{\pi}}\frac{1}{\sqrt{n}} \\
&< \frac{4}{\sqrt{n}}.
\end{aligned}$$

Inequality (i) follows from Equation (12), and Inequality (j) follows from $1 + 1/n \leq 3/2$ when $n \geq 2$ and Lemma 3.

Thus, we have

$$\left|\Pr\left[\mathsf{DenGame}^0_{\mathcal{A}} = 1\right] - \Pr\left[\mathsf{DenGame}^1_{\mathcal{A}} = 1\right]\right| \leq \frac{4}{\sqrt{n}},$$

and if we let $n = 16\delta(\lambda)^2$, the DFHE scheme $\mathsf{DFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec}, \mathsf{Fake})$ is $1/\delta(\lambda)$-deniable. □

4.2 Efficiency Analysis

According to Equations (3) and (4), the bit length of the randomness is $(1 + (1-c)k'\ell'_c + ck'\ell)n$ and the number of bootstrapping operations in the encryption algorithm is $(2k'+1)n - 1$. Therefore, when n is reduced from $100\delta(\lambda)^2$ to $16\delta(\lambda)^2$, the two values are reduced by 84%.

5 Faster DFHE Using TFHE

In FHE, the execution time of bootstrapping is extremely long. Accelerating bootstrapping is crucial for the practical applications of FHE. The DFHE scheme proposed by Agrawal et al., as described in Construction 1, also employs bootstrapping. Employing an FHE scheme with a faster bootstrapping than the BGV scheme [6] employed by Agrawal et al. can lead to a faster DFHE scheme. Therefore, we employ TFHE [11], known for its fast bootstrapping.

TFHE is a concrete FHE construction proposed by Chillotti et al. [11]. While the original TFHE proposed by Chillotti et al. is a symmetric-key FHE, Agrawal et al.'s DFHE construction requires a public-key FHE. Hence, we employ a public-key variant of TFHE. Several public-key TFHE schemes have been proposed by Joye [19,20]. We employ the scheme proposed in [20, Section 3] due to its smaller public key size and smaller noise variance.

We recall Joye's public-key TFHE scheme [20, Section 3] below. According to the functional completeness of the NAND gate [25], which states that any logic circuit can be constructed solely from NAND gates, we describe $\mathsf{TFHE.Eval}$ below specifically for homomorphic NAND computation.

Construction 2 (TFHE [20]) *Let $\circledast\colon \mathbb{Z}^n \times \mathbb{Z}^n \to \mathbb{Z}^n$ be a reverse negative wrapped convolution.*[3] *A public-key TFHE scheme for the message space $\mathcal{M} = \{0,1\}$, $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$, is described below.*

– $\mathsf{TFHE.Gen}(1^\lambda) \to (\mathsf{pp}, \mathsf{pk}, \mathsf{sk})$*: On input the unary representation of the security parameter λ, do the following:*
  1. *Set $n = \varphi(M)$ for some $M \in \mathbb{N}$.*
  2. *Select $q \in \mathbb{Z}_{\geq 0}$ with $2|q$ and set $\Delta = q/2$.*
  3. *Set two discretized error distributions χ_1 and χ_2 over $\mathbb{Z}$ with a variance σ.*
  4. *Set $\widetilde{\boldsymbol{b}} = \widetilde{\boldsymbol{a}} \circledast \boldsymbol{s} + \boldsymbol{e}$ with $\boldsymbol{s} = (s_1, \ldots, s_n) \xleftarrow{\$} \{0,1\}^n$, $\widetilde{\boldsymbol{a}} = (\widetilde{a}_1, \ldots, \widetilde{a}_n) \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^n$, and $\boldsymbol{e} \leftarrow \chi_1^n$.*
  5. *Output $\mathsf{pp} = (n, \sigma, 2, q, \Delta)$, $\mathsf{pk} = (\widetilde{\boldsymbol{a}}, \widetilde{\boldsymbol{b}})$, and $\mathsf{sk} = \boldsymbol{s}$.*

– $\mathsf{TFHE.Enc}(\mathsf{pk}, m) \to \mathsf{ct}$*: On input a public key pk and a message m, do the following:*
  1. *Set $\boldsymbol{a} = \widetilde{\boldsymbol{a}} \circledast \boldsymbol{r} + \boldsymbol{e}_1$ and $b = \langle \widetilde{\boldsymbol{b}}, \boldsymbol{r} \rangle + \Delta m + e_2$ with $\boldsymbol{r} \xleftarrow{\$} \{0,1\}^n$, $\boldsymbol{e}_1 \leftarrow \widehat{\chi}_1^n$, and $e_2 \leftarrow \widehat{\chi}_2$.*
  2. *Output $\mathsf{ct} = (\boldsymbol{a}, b)$.*

– $\mathsf{TFHE.Eval}((\mathsf{BRK}, \mathsf{KSK}), \mathsf{NAND}, \mathsf{ct}_1, \mathsf{ct}_2) \to \mathsf{ct}$*: On input a blind rotation key BRK, a key switching key KSK, a circuit for NAND computation, and two ciphertexts ct_1 and ct_2, do the following:*[4]
  1. *Output $\mathsf{ct} = \mathsf{boot}((\mathsf{BRK}, \mathsf{KSK}), (\boldsymbol{0}, 5q/8) - \mathsf{ct}_1 - \mathsf{ct}_2)$.*

– $\mathsf{TFHE.Dec}(\mathsf{sk}, \mathsf{ct}) \to m$*: On input a secret key sk and a ciphertext ct, do the following:*
  1. *Output $\lfloor((b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q)/\Delta\rceil \bmod 2$.*

To show that the TFHE scheme (Construction 2) can be transformed into a DFHE scheme using Agrawal et al.'s construction (Construction 1), we prove the following Theorem 2.

Theorem 2. *The public-key TFHE scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ described in Construction 2 is a weak version of special FHE scheme.*

Proof. We prove that the FHE scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies the five properties of special FHE (Definition 5).

1. The $\mathsf{TFHE.Eval}$ is deterministic because its underlying boot is deterministic when the keys BRK and KSK are fixed (see [11,18] for details on bootstrapping). The $\mathsf{TFHE.Dec}$ is also deterministic. Hence, the scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies property 1 of special FHE.
2. The scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies property 2 of special FHE, assuming the hardness of the learning with errors (LWE) problem [22].
3. For all $\mathsf{ct} = (\boldsymbol{a}, b) \in (\mathbb{Z}/q\mathbb{Z})^{n+1}$ and all $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{TFHE.Gen}(1^\lambda)$, we have
$$\mathsf{TFHE.Dec}(\mathsf{sk}, \mathsf{ct}) = \left\lfloor \frac{(b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q}{\Delta} \right\rceil \bmod 2 \in \{0,1\} = \mathcal{M}.$$
Hence, the scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies property 3 of special FHE.
4. We have
$$\begin{aligned}
\Pr[\mathsf{TFHE.Dec}(\mathsf{sk}, \mathsf{ct}) = 0] &= \Pr\left[\left\lfloor \frac{(b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q}{\Delta} \right\rceil \bmod 2 = 0\right] \\
&\overset{\text{(a)}}{=} \sum_{i=-\infty}^{\infty} \Pr\left[\left\lceil \frac{4\{(b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q\} - q}{2q} \right\rceil = 2i\right] \\
&\overset{\text{(b)}}{=} \sum_{i=-\infty}^{\infty} \Pr\left[2i - 1 < \frac{4\{(b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q\} - q}{2q} \leq 2i\right] \\
&= \sum_{i=-\infty}^{\infty} \Pr\left[\frac{4i-1}{4} q < (b - \langle \boldsymbol{a}, \boldsymbol{s}\rangle) \bmod q \leq \frac{4i+1}{4} q\right]
\end{aligned}$$

[3] For the formal definition of reverse negative wrapped convolution [20], see Definition 13 in Appendix A.3.

[4] Although we omitted them from this paper for simplicity, the blind rotation key BRK and the key switching key KSK are the bootstrapping keys generated by the $\mathsf{TFHE.Gen}$ algorithm. These keys are made public as part of the public key.

$$\stackrel{\text{(c)}}{=} \Pr\left[0 \le (b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q \le \frac{1}{4}q\right] + \Pr\left[\frac{3}{4}q < (b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q \le q-1\right], \tag{13}$$

where $\mathsf{ct} = (\boldsymbol{a}, b) \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^{n+1}$ and $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{TFHE.Gen}(1^\lambda)$. Equation (a) follows from $\Delta = q/2$, $\lfloor x \rceil = \lceil x - 1/2 \rceil$ for all $x \in \mathbb{R}$, and a property of the modulo-2 operation. Equation (b) follows from the definition of the ceiling function. Equation (c) follows from $\Pr\left[\frac{4i-1}{4}q < (b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q \le \frac{4i+1}{4}q\right] = 0$ for all $i \in \mathbb{Z} \setminus \{0, 1\}$ and a property of the modulo-q operation.
Since $(\boldsymbol{a}, b) \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^{n+1}$, we have

$$\Pr[b = 0] = \cdots = \Pr[b = q-1] = \frac{1}{q}. \tag{14}$$

In addition, for all $j \in [0, q-1]$, let $p(j) \coloneqq \Pr[\langle \boldsymbol{a}, \boldsymbol{s} \rangle \bmod q = j]$. Then, we have

$$\sum_{j=0}^{q-1} p(k) = 1. \tag{15}$$

Hence, for all $k \in [0, q-1]$, we have

$$\begin{aligned}
&\Pr[(b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q = k] \\
&= \Pr[b = q-1]p(q-1-k) + \cdots + \Pr[b = k]p(0) + \Pr[b = k-1]p(q-1) + \cdots + \Pr[b = 0]p(q-k) \\
&\stackrel{\text{(d)}}{=} \frac{1}{q}\{p(q-1-k) + \cdots + p(0) + p(q-1) + \cdots + p(q-k)\} \\
&\stackrel{\text{(e)}}{=} \frac{1}{q},
\end{aligned} \tag{16}$$

where $\mathsf{ct} = (\boldsymbol{a}, b) \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^{n+1}$ and $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{TFHE.Gen}(1^\lambda)$. Equation (d) follows from Equation (14). Equation (e) follows from Equation (15).
Hence, we have

$$\begin{aligned}
\Pr[\mathsf{TFHE.Dec}(\mathsf{sk}, \mathsf{ct}) = 0] &\stackrel{\text{(f)}}{=} \sum_{k=0}^{\lfloor \frac{1}{4}q \rfloor} \Pr[(b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q = k] + \sum_{k=\lceil \frac{3}{4}q \rceil}^{q-1} \Pr[(b - \langle \boldsymbol{a}, \boldsymbol{s} \rangle) \bmod q = k] \\
&\stackrel{\text{(g)}}{=} \frac{1}{q}\left\{\left(\left\lfloor \frac{1}{4}q \right\rfloor - 0 + 1\right) + \left(q - 1 - \left\lceil \frac{3}{4}q \right\rceil + 1\right)\right\} \\
&= \frac{1}{q}\left(\left\lfloor \frac{1}{4}q \right\rfloor + 1 + q - \left\lceil \frac{3}{4}q \right\rceil\right) \\
&\stackrel{\text{(h)}}{>} \frac{1}{q}\left\{\frac{1}{4}q - 1 + 1 + q - \left(\frac{3}{4}q + 1\right)\right\} \\
&= \frac{1}{2} - \frac{1}{q},
\end{aligned}$$

where $\mathsf{ct} = (\boldsymbol{a}, b) \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^{n+1}$ and $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{TFHE.Gen}(1^\lambda)$. Equation (f) follows from Equation (13). Equation (g) follows from Equation (16). Inequality (h) follows from $x - 1 < \lfloor x \rfloor, \lceil x \rceil < x + 1$ for all $x \in \mathbb{R}$.
Thus, the scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies property 4 of special FHE.

5. TFHE assumes circular security [11,21]. Hence, the scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ satisfies property 5 of special FHE.

Thus, the scheme $\mathsf{TFHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ is a weak version of special FHE scheme. □

A single bootstrapping operation in the BGV scheme has a latency of 15 seconds, whereas in TFHE it is merely 0.0295 seconds [3]. In the DFHE construction by Agrawal et al., which relies on frequent sequential bootstrapping operations, using TFHE instead of BGV results in a significant speedup, even accounting for BGV's single instruction multiple data (SIMD) bootstrapping operations.

6 Conclusion

In this work, we improved the efficiency of the DFHE scheme by Agrawal et al. using two techniques. First, we reduced the bit length of the randomness and the number of bootstrapping operations in the encryption algorithm by deriving a tighter upper bound on the scheme's deniability. Second, we constructed a faster DFHE scheme by showing that TFHE, which is known for its fast bootstrapping, is a special FHE.

Future work includes investigating constructions of DFHE based on other FHE schemes and further improving the algorithm's efficiency. Our research did not achieve an asymptotic (i.e., order-level) improvement in performance. For real-world applications, achieving such an order-level efficiency improvement is a crucial next step.

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A Additional Preliminaries

A.1 Omitted Definitions for FHE

Definition 6 (Correctness). *An FHE scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *is correct if for all security parameters* λ*, all polynomial-time circuits* $C\colon \mathcal{M}^k \to \mathcal{M}$*, and all messages* $m_i \in \mathcal{M}$ *for* $i \in [k]$*, we have*

$$\Pr\left[\begin{matrix}\mathsf{Dec}(\mathsf{sk}, \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)) \\ = C(m_1, \ldots, m_k)\end{matrix} : \begin{matrix}(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda) \\ \forall i \in [k], \mathsf{ct}_i \leftarrow \mathsf{Enc}(\mathsf{pk}, m_i)\end{matrix}\right] \geq 1 - \mathsf{negl}(\lambda).$$

Definition 7 (Compactness). *An FHE scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *is compact if there exists a polynomial* $\mathsf{poly}(\cdot)$ *such that for all security parameters* λ*, all polynomial-time circuits* $C\colon \mathcal{M}^k \to \mathcal{M}$*, and all messages* $m_i \in \mathcal{M}$ *for* $i \in [k]$*, we have*

$$\Pr\left[|\mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)| \leq \mathsf{poly}(\lambda) : \begin{matrix}(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda) \\ \forall i \in [k], \mathsf{ct}_i \leftarrow \mathsf{Enc}(\mathsf{pk}, m_i)\end{matrix}\right] = 1.$$

Definition 8 (IND-CPA Security). *An FHE scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *is IND-CPA secure if for all PPT adversaries* $\mathcal{A}$*, we have*

$$\left|\Pr\left[\mathsf{CPAGame}^0_{\mathcal{A}}(\lambda) = 1\right] - \Pr\left[\mathsf{CPAGame}^1_{\mathcal{A}}(\lambda) = 1\right]\right| \leq \mathsf{negl}(\lambda),$$

where $\mathsf{CPAGame}^b_{\mathcal{A}}(\lambda)$ *is a game between a challenger and a PPT adversary* $\mathcal{A}$ *with a challenge bit* b *defined as follows:*

- *The challenger samples* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$*, and sends* pk *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *chooses* $m_0, m_1 \in \mathcal{M}$*, and sends* (m_0, m_1) *to the challenger.*
- *The challenger computes* $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b)$*, and sends* ct *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *outputs a bit* b' *which we define as the output of the game.*

Definition 9 (Circular Security). *A public-key encryption scheme with key generation algorithm* Gen *and encryption algorithm* Enc *is circular secure if for all PPT adversaries* $\mathcal{A}$*, we have*

$$\left|\Pr\left[\mathsf{CircGame}^0_{\mathcal{A}}(\lambda) = 1\right] - \Pr\left[\mathsf{CircGame}^1_{\mathcal{A}}(\lambda) = 1\right]\right| \le \mathsf{negl}(\lambda),$$

where $\mathsf{CircGame}^b_{\mathcal{A}}(\lambda)$ *is a game between a challenger and a PPT adversary* $\mathcal{A}$ *with a challenge bit* b *defined as follows:*

- *The challenger samples* $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$*, computes* $\mathsf{ct}_{\mathsf{sk}} \leftarrow \mathsf{Enc}(\mathsf{pk}, \mathsf{sk})$*, and sends* $(\mathsf{pk}, \mathsf{ct}_{\mathsf{sk}})$ *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *chooses* $m_0, m_1 \in \mathcal{M}$*, and sends* (m_0, m_1) *to the challenger.*
- *The challenger computes* $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b)$*, and sends* ct *to* $\mathcal{A}$*.*
- *The adversary* $\mathcal{A}$ *outputs a bit* b' *which we define as the output of the game.*

Definition 10 (Valid Ciphertext). *A ciphertext* ct *of an FHE scheme* $\mathsf{FHE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ *is a valid ciphertext of* m *if either*

$$\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m),$$

or for all polynomial-sized circuits $C\colon \mathcal{M} \to \mathcal{M}$*, we have*

$$\Pr\left[\mathsf{Dec}(\mathsf{sk}, \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct})) = C(m) : (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)\right] \ge 1 - \mathsf{negl}(\lambda).$$

Definition 11 (Addition Modulo 2). *The homomorphic evaluation of addition modulo 2 circuit, denoted by* $\oplus_2\colon (\mathcal{R}^{\ell_c})^k \to \mathcal{R}^{\ell_c}$*, is defined as*

$$\oplus_2(\mathsf{ct}_1, \ldots, \mathsf{ct}_k) = \mathsf{ct},$$

where $x_i \in \{0,1\}$ *for* $i \in [k]$*,* ct_i *is a valid ciphertext of* x_i *for* $i \in [k]$*, and* ct *is a valid ciphertext of* $\sum_{i=1}^k x_i \pmod 2$*.*

A.2 Omitted Definition for DFHE

Definition 12 (Deniability Compactness [2]). *A* $1/\delta(\lambda)$*-deniable scheme* DFHE *is deniability compact if there exists a polynomial* $\mathsf{poly}(\cdot)$ *such that for all security parameters* λ *and all messages* $m \in \mathcal{M}$*, we have*

$$\Pr[|\mathsf{Enc}(\mathsf{dpk}, m)| \le \mathsf{poly}(\lambda)] = 1,$$

regardless of the encryption running time.

A.3 Omitted Definition for TFHE

Definition 13 (Reverse Negative Wrapped Convolution [20]). *For* $n \in \mathbb{N}$ *and* $i \in [n]$*, the reverse negative wrapped convolution of two vectors* $\boldsymbol{u} = (u_1, \ldots, u_n), \boldsymbol{v} = (v_1, \ldots, v_n) \in \mathbb{Z}^n$ *is the vector* $\boldsymbol{w} = \boldsymbol{u} \circledast \boldsymbol{v} = (\boldsymbol{u} \circledast_1 \boldsymbol{v}, \ldots, \boldsymbol{u} \circledast_n \boldsymbol{v}) \in \mathbb{Z}^n$ *defined by*

$$\boldsymbol{u} \circledast_i \boldsymbol{v} \coloneqq \sum_{j=1}^{i} u_j v_{n+j-i} - \sum_{j=i+1}^{n} u_j v_{j-i}.$$